

Anirudh Reddy Kunduru

Machine Learning Engineer

SUMMARY

Machine Learning Engineer with **5+ years** of experience delivering production-ready ML systems across cloud and distributed environments. Strong in **deep learning, sequence modeling, reinforcement learning**, and hands-on development with **Python, PyTorch, TensorFlow**, and **scikit-learn**. Skilled in **AWS, Azure, Kafka, Flink**, and end-to-end **MLOps** using **MLflow, Airflow, Docker**, and **Kubernetes**. Experienced in building **real-time inference pipelines**, optimizing **latency** and **throughput**, and designing **scalable architectures** that support predictive modeling, **recommendation engines**, and high-impact business applications.

TECHNICAL SKILLS

Programming & Scripting: Python (NumPy, Pandas, scikit-learn, TensorFlow, PyTorch), R, Scala

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs), NLP (BERT, spaCy, Transformers), Time Series Forecasting, Feature Engineering, Ensemble Methods (XGBoost, Random Forest), Reinforcement Learning, Hyperparameter Optimization, Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG)

Cloud Platforms & Warehousing: AWS (EC2, S3, Glue, Redshift, SageMaker, Lambda), Azure (Azure Data Lake, Azure ML, AKS), Snowflake

MLOps & Deployment: Docker, Kubernetes, Kubeflow, MLflow, Flask, FastAPI, Jenkins, Terraform, CI/CD, Model Monitoring, Model Versioning

Data Engineering & Big Data: Apache Spark, Hadoop, Kafka, Airflow, AWS EMR, Data Lakes, Prefect, SQL (MySQL, PostgreSQL), NoSQL

Data Analysis & Visualization: EDA, A/B Testing, Statistical Analysis, Predictive Analytics, Data Wrangling, Tableau, Power BI, SSRS

Tools & Environments: Jupyter Notebook, RStudio, Linux Shell, VS Code, ELK Stack, Git/GitHub, RESTful Services

PROFESSIONAL EXPERIENCE

Netflix | *Machine Learning Engineer*

CA | January 2025 – Present

- Designed and deployed a foundation-model-based **sequence prediction system** using **Transformer architectures** on **AWS SageMaker** and **EC2 GPU clusters**, improving 6-month member retention by **7%** through more accurate long-term viewing **behavior modeling**.
- Optimized real-time inference performance in **PyTorch** and **TensorFlow Serving** by implementing **key-value caching**, **model quantization**, and distributed deployment on **AWS EKS**, reducing latency by **30%** and supporting **2M+ requests/sec** globally.
- Developed a **Reinforcement Learning** reward **optimization framework** using **AWS Lambda** and **Step Functions** to balance **short-term engagement** with **long-term subscriber LTV**, delivering a **5% increase** in **satisfaction** and **retention metrics**.
- Built **real-time feature pipelines** with **Apache Kafka** and **Apache Flink** to process high-volume clickstream data, enabling sub-second model refreshes and improving **real-time recommendation relevance** by **12%**.
- Supported the design and upkeep of **MLOps pipelines** using **Airflow**, **MLflow**, **Docker**, and **Kubernetes** to streamline **model training**, tracking, and deployments, helping improve reliability and speed of production releases.
- Collaborated with **data** and **infra teams** to **develop vector-based retrieval systems** using **FAISS** and **Pinecone**, enhancing embedding-driven **candidate generation** for recommendation models and boosting **cold-start accuracy** by **10%**.

Accenture | *Machine Learning Engineer*

India | July 2020 – December 2023

- Engineered and productionized **end-to-end machine learning solutions** across **finance** and **retail**, translating business requirements into scalable **ML pipelines** using **Python**, **scikit-learn**, and **TensorFlow**.
- Built **cloud-native ML infrastructure** on **Microsoft Azure** (Azure Data Lake, Azure ML, AKS) to automate data ingestion, model training, and deployment for **20+ predictive models**, reducing infrastructure costs by **12%** and accelerating model rollout by **20%**.
- Developed **RESTful APIs** with **Flask** and **FastAPI** to operationalize ML models, lowering inference latency by **18%** and improving integration with **microservices** and **client-facing applications**.
- Deployed optimized **CNN** and **RNN models** for **computer vision** and **time-series forecasting workloads**, improving predictive accuracy by **15%**, reducing false positives by **10%**, and increasing inference speed by **15%**.
- Implemented robust **MLOps** and **CI/CD** pipelines using **Azure DevOps**, **MLflow**, and **Git** to enable continuous integration, automated retraining, reproducibility, and model drift monitoring across production environments.
- Created **interactive Power BI dashboards** to track **ML performance metrics**, business **KPIs**, and **model ROI**, enabling stakeholders to make informed, data-driven decisions.

EDUCATION

Master of Science in Computer Science | University of Central Missouri, MO, USA.

Bachelor of Science in Computer Science | Anurag University, Hyderabad, India.